

FULLY DIFFERENTIAL GAIN-BOOSTED OTA DESIGN FOR PIPELINE DA-CONVERTER IN 180NM CMOS TECHNOLOGY

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ABSTRACT

This paper describes a fully differential gain-boosted improved OTA for high performance pipeline DA-converter [1], [2] in UMC 180nm technology for a 3.3V power supply applications. The architecture chosen for the design is a folded cascode OTA structure for the main and gain boosting stages. The designed OTA achieved a very high DC gain greater than 95db with a unity gain band width of 500MHz for a 500fF load with a power consumption of 14.5mW and phase margin of more then 55 degrees, which meets the design specifications. The design is meant to be used for implementation of a pipeline DA-converter, using selection inversion based on Gray coded bit vector. Simulation is done using Cadence Affirm Spectre circuit simulator. Layout of the OTA structure has been finished and prepared to be taped out.

1. INTRODUCTION

With the scaling down of technology, design of analog CMOS functional blocks with low power low voltage regime becomes more and more difficult due to the deviation from the long channel characteristics. The contradictory demand of speed and accuracy make the analog and mixed signal design far more difficult. The speed and accuracy of the operational amplifier or operational transconductance amplifier (OTA) determine the settling behavior of a wide variety of analog circuit such as AD-DA converters, sigma-delta converters, filter, S/H circuit. Fast settling of the system requires a high unity gain frequency and a single-pole roll off, whereas accurate settling requires a very high DC gain [3].

Technology scaling increases the device intrinsic unity gain frequency [4] and hence increases the system level speed, but the main drawback is the reduction of DC gain of the gain stage. The gain boosting technique can combat the reduction of DC gain without any reduction of bandwidth [3], [5], [6]. The technique decouples the DC gain and frequency response of the OTA. Fully differential gain boosting can reduce the circuit complexity as well as reduce the power consumption.

The single-ended gain boosting technique in Figure 1 uses auxiliary four OTAs with the main OTA to enhance the DC gain. This increases the signal path and introduces an extra pole from an internal current mirror; therefore it has poor frequency response and needs careful design methods described in [3], [10], [11] to follow to alleviate the performance. Circuit complexity and more transistors will introduce more noise, which in turn reduces the dynamic range of the OTA.

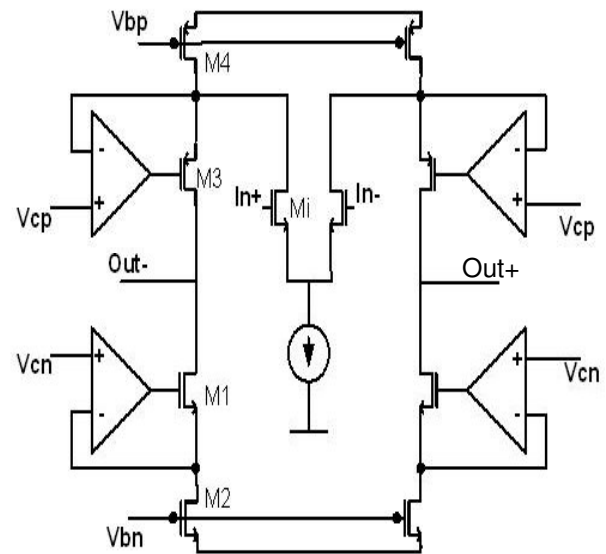


Figure 1. Single-ended gain boosted folded cascode OTA.

The problems are addressed by using fully differential gain-boosting techniques Figure 2. with high potential to produce better system level performance as well. This improves the settling behavior and reduces noise with less number of transistors as well. The fully differential single stage improved OTA design has two auxiliary OTAs with the main OTA.

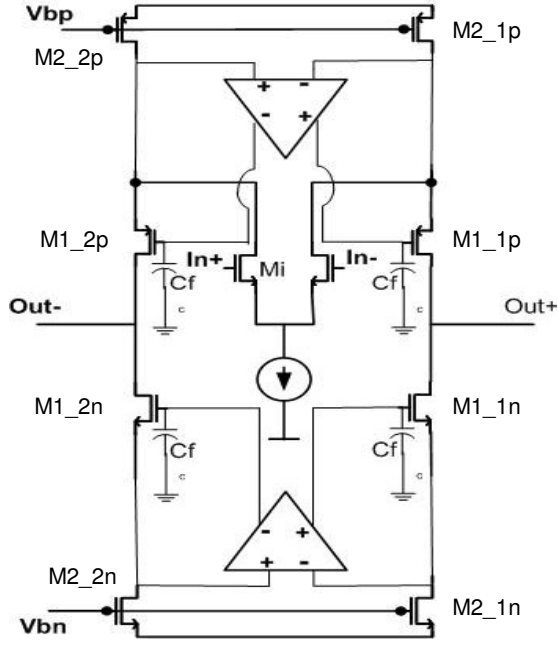


Figure 2. Fully differential gain boosted folded cascade OTA

2. THE OTA DESIGN

2.1. Selection of OTA architecture

The on-chip amplifier needs to drive capacitive load with high speed, so we prefer to use a single stage OTA rather than 2-stage operational amplifier. Even though the telescopic OTA architecture has the advantage of low power consumption but lower output swing range makes it less preferable in low voltage applications. The folded cascode architecture with fully differential gain boosting is chosen as the optimum structure for this design. It has relatively large output signal swing and good single pole roll-off frequency response with a simple design structure. Additionally, it has relatively high common mode input range.

2.2. Principle of gain-boosting

The technique of gain boosting is based on increasing the cascoding effect of load transistor by adding an additional gain stage. This also reduces the feedback from output to the input transistors [3]. Thus the output impedance of the main stage in Figure 1 or 2 is increased by the gain of the additional cascoded stage.

$$R_{out} = \frac{1}{2}(g_{m2} r_{o2} (A_{add} + 1) + 1) r_{o1} + r_{o2} \quad (1)$$

Where, g_{m2} , is the transconductance of the transistor closer to the output node. r_{o2} and r_{o1} are the output impedance of the transistors near the output node and the $\frac{1}{2}$ term is due to the parallel combination of AC

impedance branch from output node to V_{ss} and V_{dd} . A_{add} is the open loop gain of the additional gain stages.

This in turn increases the total DC gain of the circuit. Assuming $r_{o1} = r_{o2} = r_o$ and $g_{m1} = g_{m2} = g_m$ we can also approximate the gain as in Equation 3:

$$A_{o,tot} = -g_{m1} R_{out} \quad (2)$$

$$A_{o,tot} \approx -\frac{1}{2}(g_m r_o)^2 A_{add} \approx -\frac{1}{4}(g_m r_o)^4 \quad (3)$$

If exactly the same OTA circuit is used for the boosting stage then the overall gain can be approximated as 12 dB less than that of $(g_m r_o)^4$.

It has been reported that additional stage need not be fast as the main stage, also for first order roll-off $(f_{3dB})_{total}$ have to be smaller than $(f_{3dB})_{add}$ and,

$$(f_{3dB})_{main} < (f_u)_{add} < (f_{2ndpole})_{main} \quad (4)$$

must be maintained for stable operation of the OTA structure [3], [10]. Illustrated in Figure 3.

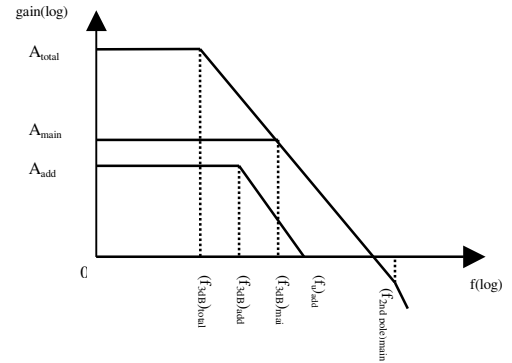


Figure 3. Frequency response design constrain for gain boosted OTA design

All the design constrains are carefully maintained in the final design.

2.3. Design steps

The transistors are biased in the saturation region to get high output resistance. The output branch current is made half of the input branch current to make the input g_m and output resistance higher so that they produce high DC-gain and reduce thermal noise of the input pair [5-6]. One disadvantage of doing this is to reduce the slew rate. To improve the slew rate, transistor M12 and M13 in Figure 4, with moderate size are used in each stage [7]. They also prevent the high voltage swing of the folded cascode node without any additional loading effect.

The main stage OTA architecture is an n-channel input stage. The top gain boosting OTA stage in Figure 2 drives the PMOS load is also a NMOS input pair folded cascade, which is biased to give highest gain with a common mode input voltage equal to one overdrive voltage less than the supply voltage and having a common mode output drive voltage which will produce the desired AC ground for the main block i.e. half of the supply voltage.

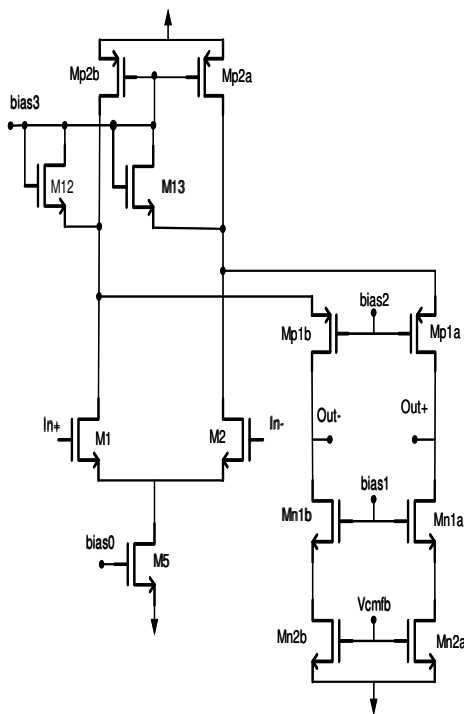


Figure 4. Fully differential folded cascode OTA architecture with slew-rate improving technique.

The bottom gain boosting OTA stage in Figure 2 is PMOS input stage folded cascode structure with one overdrive higher than input common mode from the DC ground and output common mode voltage is maintained for the same requirement of the above circuit. Extra transistors M12 and M13 can be added to improve the slew rate without affecting the AC performance of the main OTA structure. The gain boosting OTA structures are not slew rate limiting. Extra care can be taken to minimize the input transistor widths of the gain boosting OTAs so that they don't overload the intermediate node of the main OTA. This design consideration is very important to push the 2nd pole of the AC response as high as possible.

One pole-zero doublet will be present in the final output of the OTA, near the unity gain frequency [6]. This may degrade the settling behaviour of the circuit. Careful selection of capacitor Cf can introduce necessary

feedback to improve the settling behaviour by pushing the pole-zero doublet to higher frequency.

Proper switched-capacitor common mode feedback circuit [4], [7] has been added for all the three OTAs to maintain the common mode output voltages. A biasing circuit has to be designed to operate the circuit at the desired operation point.

The Layout of the design has been finished for UMC180 nm CMOS technology using Virtoso Layout XL from Cadence.

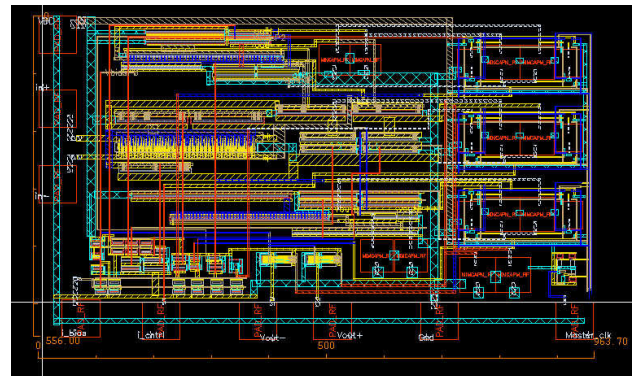


Figure 5. Layout of complete OTA structure with biasing circuit and SC CMFB circuit (400um * 800um without the measuring RF-PAD structure)

3. RESULTS

The simulation for the OTA circuits is done with a 500fF load, which is a close approximation for the loaded OTA in the pipelined DA-converter circuit. AC simulation has been done for single ended output.

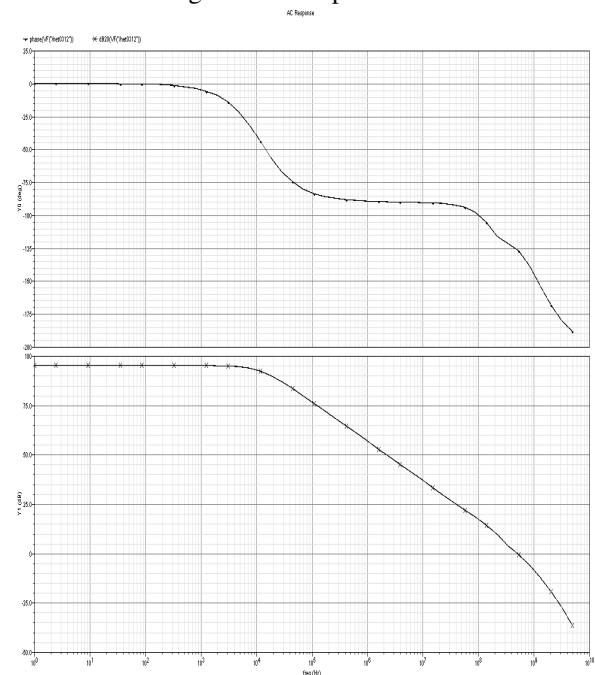


Figure 6. Frequency response of the OTA.

The summary of the OTA performance is given below,
DC gain = 95dB,

Phase margin = 55°
 Slew rate = $420\text{V}/\mu\text{s}$
 Unity gain BW = 500MHz
 Power consumption = 14.5mW

The performance can be adjusted at the cost of increase in power consumption, depending on the requirement of the application. If we increase the current through proper adjustment of the W/L ratio then we can easily increase the BW of the OTA as well as it will increase the slew rate as in Figure 7 [9]. It can be a better choice to make the input and output branch current equal for high frequency operation which will also increase the slew rate of the circuit in turns with little decrease of the DC gain.

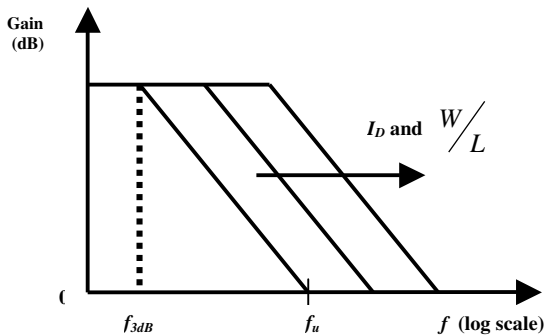


Figure 7. Increase of BW through increase current and larger transistor aspect ratio.

4. CONCLUSION

The technique used in this design actually decouples the DC gain and the unity gain bandwidth with some additional method to improve the degraded slew-rate. Many design issues are there to cope with the design criteria and the trade off between the parameters. The system level simulation for the DA converter is in progress. As the DA converter is a pipelined design the predicted accurate settling behaviour due to the very high DC gain will provide some potential to run the system with lower frequency and large number of pipeline stages. The designed IP-core of the high DC-gain OTA is ready for tape out with some test circuitry.

5. ACKNOWLEDGMENT

This work was supported by the SSF-funded program Radio and Mixed Signal Circuit and System Design (RaMSiS).

6. REFERENCES

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